

Beyond Centrality: Revealing Disease-Critical Proteins in a COVID-19 Interactome Through Minimum Vertex Cuts

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COVID-19 became a leading cause of death worldwide within months of its emergence in late 2019, and severe disease continues to kill patients despite widespread vaccination and antiviral therapy. What distinguishes fatal COVID-19 from mild infection is not the virus alone but a coordinated failure of host systems — immune regulation, coagulation, protein quality control, and cell-cycle regulation — that together drive the progression from infection to organ failure. Despite years of intense investigation, no consensus exists on which proteins occupy the structural bottlenecks through which viral activation signals propagate to produce clinical pathology. Centrality-based analyses of the interactome rank proteins by local connectivity, but a ranked list is not a bottleneck: no centrality measure identifies the smallest collective set of proteins whose removal disconnects disease initiation from clinical outcome.

The protein interactome of SARS-CoV-2 infection — the web of physical and regulatory interactions through which the virus hijacks host biology — is, by definition, a directed capacitated graph: proteins are nodes, interactions are arcs, and differential expression between infected and control patients provides a natural capacity on the throughput of each node. No prior work has formulated the identification of disease-critical proteins in COVID-19 as a minimum vertex cut on this interactome. This project does so, using a network I constructed from 8,062 proteomic features measured in 202 patients (102 COVID-19 positive, 100 controls) at a major academic medical center.

The network. I built a directed interactome of 18,832 nodes and 75,972 edges by integrating five curated databases (IntAct, STRING, BioGRID, Reactome, KEGG) with a patient-derived co-expression matrix computed across the 202-patient cohort. Undirected edges were oriented using a signal-propagation algorithm that directs flow from low-degree peripheral nodes toward high-degree hubs. The resulting interactome has a giant connected component of 12,897 proteins with graph diameter 6, and 119 connected components in total.

Problem formulation — disease-critical proteins as a minimum vertex cut. Source and sink node sets are defined by the biology of the disease process: sources are proteins mediating viral entry and initial immune activation (ACE2, TMPRSS2, complement initiators), and sinks are terminal effectors of clinical pathology (fibrin formation, inflammatory cytokine release). Each

internal protein v is replaced by v_{in} and v_{out} connected by an internal arc whose capacity is derived from differential expression between COVID-19 and control patients; all original arcs receive infinite capacity (Transformation T1, AMO Chapter 8). Maximum flow from source to sink is computed via the Preflow-Push algorithm (AMO Chapter 7), and by the max-flow/min-cut theorem (AMO Chapter 6) the saturated internal arcs identify the minimum vertex cut — the smallest set of proteins whose removal disconnects viral activation from clinical pathology. Minimality is certified exactly by Menger’s theorem, which additionally yields the maximum number of node-disjoint source–sink paths as the dual quantity.

Why this matters. This formulation resolves a striking graph-theoretic paradox I encountered in prior analysis of the same dataset. The proteins that best separate patient groups in a classification sense (LUM, PAGR1, CCNB1, CDC6) are low-degree leaf nodes, connected to only a handful of neighbors. The highest-degree hub proteins (UBC, HSPA8, HSPA5), which participate in hundreds of interactions, carry nearly identical values in the two groups and almost no discriminative power. The best classifiers are structurally peripheral; the structurally central nodes are poor classifiers. Neither set answers the underlying structural question — which proteins form the minimum bottleneck between disease input and disease output. The minimum vertex cut identifies a third category of flow-theoretic bottleneck proteins, certified by Menger’s theorem and independent of both classifier performance and degree centrality.

Deliverables. The project will be implemented in Python using NetworkX and CVXPY. Deliverables include the maximum-flow solution and its induced minimum vertex cut on the transformed interactome of approximately 37,700 nodes, with its certificate of minimality and the Menger dual (the maximum number of node-disjoint source–sink paths, establishing the vertex connectivity between disease initiation and clinical pathology); an expression-weighted version in which patient-derived differential expression calibrates the internal arc capacities; and a three-way comparison of the cut-derived protein panel against the highest-degree hub proteins and the statistical classifiers identified in prior analysis, with bootstrap validation and sensitivity analysis under alternative source/sink definitions. These results will demonstrate that a biomedical question traditionally addressed by statistics and machine learning admits a natural network-flow formulation whose solution carries exact mathematical certification — and points to a new category of disease-critical proteins, invisible to both classifier-based and centrality-based analysis, that may warrant further investigation as intervention targets in severe COVID-19.